

GENERAL EDUCATION: SCIENCE AND TECHNOLOGY

A. General Concepts and Historical Development

1. History of S&T in the World

1.1. Ancient Three-Age System - a system of classifying ancient ages into groups based on tools developmental stages.

a. Stone Age

- Weapons made of stone, wood, bone or some other materials for hunting.
- Neanderthals and Denisovans human species

a.1. Paleolithic Period – Old Stone Age (2.5 MYA to 10,000 B.C.)

- People are mainly hunters and gatherers of food
- Living nomadically in caves
- Used basic stone tools for hunting
- Controlled fire

a.2. Mesolithic Period – Middle Stone Age (10,000 B.C. to 8,000 B.C.)

- People are still hunters and gatherers of food
- Living as permanent settlers in villages (near rivers)
- Used small stone tools for hunting
- polished, pointed spears/arrows
- Learned fishing and introduced agriculture

a.3. Neolithic Period – New Stone Age (8,000 B.C. to 3,000 B.C.)

- People are using agriculture and domesticate animals for food
- Living as permanent settlers in villages (near rivers)
- Used stone tools in agriculture
- Polished hand axes
- Advanced farming, home construction, and art

b. Bronze Age (3,000 B.C. to 1,300 B.C.)

- Tools and weapons were widely made with copper and bronze.
- Smelting is a process of extracting metal from involving heating and melting.
- Sumerians discovered that a harder and stronger metal could be made by blending copper and tin.
- Different human societies entered the Bronze Age at different times.
- Rise of states or kingdoms—large-scale societies joined under a central government by a powerful ruler.
 - Sumer and Babylonia in Mesopotamia
 - Athens in Ancient Greece
- The Bronze Age ended around 1200 B.C. when humans began to forge an even stronger metal: iron.

c. Iron Age

- Between 1200 to 600 B.C. depending on the region
- Hittites (Turkey) firstly made steel, a much harder metal, by heating iron with carbon.
- Launched the beginning of mass production and created major advancements in warfare.
- Persians, first civilization to develop an armored cavalry completely covered in steel armor.

d. Antiquity (600 BCE to 529 CE)

- Rise of Greek civilization
- Natural philosophers – first scientists

- Pre-Socratic Philosophers
 - Thales - Assumed Earth to be floating in water
 - Anaximander - Formulated a theory of the origin and evolution of life.
 - Anaximenes - Suggests that air is the primary substance
 - Heraclitus - “change is the essence of all being” and fire plays a crucial role in the process.
 - Leucippus - Introduces the first idea of the atom, an indivisible unit of matter.
 - Democritus - thought that atoms are solid, indestructible particles that are separated by empty space
 - Pythagoras - Recognizes that Earth is sphere
 - Pythagoreans - viewed the universe as form and number.
 - Hippocrates - Recognized as Father of Medicine
 - Hippocratic Oath - encourages separation of medicine from religion
- Socratic Philosophers
 - Socrates - Contribute knowledge through dialogues using the Socratic Method
 - Plato - Coined the term element and established the Academy
 - Used abstract geometry models rather than empirical observations
 - Aristotle - Elements undergo changes when they combine and have qualities that are based on observations using our senses.
 - Proposed the fifth element as aether
 - Established the Lyceum
 - Considered as Father of Biology
 - Introduced the inductive method
- Aristarchus – originally proposed Sun-centered universe (Heliocentrism).
- Claudius Ptolemy - calculated size of Earth and its distance to the Moon
 - Earth-centered model (Geocentrism) and became the most common cosmological view until Middle Ages.
- Herophilus – Father of Anatomy
- Erasistratus – founder of Physiology
- Euclid – Father of Modern Geometry
- Archimedes – Father of Mathematics

1.2. Middle Ages (476 – 1450 A.D.)

- Also known as Medieval Period
- European history between the fall of Roman Empire and beginning of Renaissance
- Birth of ideas - many scientific discoveries and technological advancement.

a. Early/Dark Middle Ages Discoveries (476-1000 A.D.)

- Slow progress of Science in Europe, rise of Catholic Church and suppressing natural scientists' discoveries
- Islamic Empire became the most advanced civilization
- Agricultural and Transportation discoveries
 - Heavy plough
 - Horse collar
- Science in China
 - Gun powder
 - Paper
 - Mechanical clock
 - Compass - a magnetized needle placed on a straw floating on water for reliable navigation
- Science and Mathematics in India
 - Recognition of zero in mathematics
 - Introduction of decimal system

- Arab Science
 - Many of the works of the ancients have been preserved (including Ptolemy's Astronomy) because they were translated into Arabic.
 - Toledan Tables (by al-Zarkali), a collection of mathematical tables used to predict the movements of the Sun, Moon and planets relative to the fixed stars.
 - Discovery of borax
 - Firstly to synthesize sal-ammoniac (ammonium chloride)
 - Science of anatomy did not progress because dissection of corpses is not allowed by Islamic law.

b. High Middle (1000-1450 A.D.)

- Rebirth of Science in Europe
- St. Thomas Aquinas – founder of Scholastic Schools
 - Scholastic – Christian philosophers that set out to absorb the newly gained knowledge of the ancients (Plato and Aristotle) and reconcile it with teachings of the church
- Roger Bacon – use experimentation to gain new knowledge
- Many scientists started to question Aristotelian teachings (Theory of Motion and Earth's fixity)
- Water Wheel - most important source of mechanical power throughout northern Europe for many centuries.
- Black Death - 2nd Bubonic Plague Pandemic
 - caused by the plague bacterium (*Yersinia pestis*) carried by fleas living on the black rats

1.3. Modern Age (1453 - 1659 AD)

a. Renaissance Period - scientists began to perform many experiments

- Scientific Method – scientific process based on observation and experimentation, popularized by Francis Bacon.
- Printing Press – introduced by Johannes Gutenberg using movable type printing
- Bible – world's most printed book.
- Philippus Aureolus Paracelsus – discovered Laudanum, tincture of opium as painkiller
- Tartaglia (Nicolo Fontana) – discovered Pascal's triangle and elevation of 45° for maximum distance shot of a canon
- Leonardo da Vinci – known for his engineering of canal locks, cathedrals, and engines of war.
- He was also known for of physical concepts such as inertia, and sketches for working parachutes and helicopters.

b. Scientific Revolution

- Nicolas Copernicus – successfully proposed the Heliocentric Theory, published in his book *De revolutionibus orbium caelestium* (The Revolution of Celestial Bodies)
- Johannes Kepler – Kepler's Law of Planetary Motion:
 - describe a planet's orbit as elliptical
 - the speed at which it travels
 - time it needs to complete one revolution around the Sun
- Galileo Galilei – first scientist of Scientific Revolution
 - Law of Motion (force causes acceleration)
 - Discovered pendulum and thermometer
 - Dialogue on Two New Sciences, last published work
 - Validated Copernican system using his own telescope and observations
 - craters and mountains of the moon
 - moons of Jupiter
 - phases of Venus
 - stars of Milky Way

- sunspots
- Tycho Brahe – first to describe 1572 Supernova and published the positions of 777 stars
 - determined the distance of 1577 comet from Earth using parallax
 - Believes that Sun and Moon revolved around Earth and other planets (Geo-heliocentric)
- John Ray – first to use species as the fundamental unit of classification
- Carolus Linnaeus – Father of Modern Taxonomy
 - Modern system of classification of living organisms
 - Binomial System of Nomenclature
- Andreas Vesalius – Father of Modern Anatomy
 - De humani corporis fabrica (Structure of the Human Body), the first accurate illustrations of internal human anatomy.
- William Harvey - the first to recognize the full circulation of the blood in the human body
- Marcello Malpighi discovered capillaries
- Blaise Pascal – discovered Pascaline, first mechanical calculator that used gears
- Evangelista Torricelli – invented first barometer using mercury, leading to production of first vacuum known to science

c. Enlightenment (1735-1819)

- Isaac Newton – formulated the laws of motion and universal gravitation
 - Nature of white light
 - Introduced calculus together with Gottfried Wilhelm Leibniz
- César-François Cassini – first national geographic survey, resulting in the first map produced according to modern principles.
- James Watt – steam engines
- Jean Baptiste de Lamarck – first to propose Theory of Evolution he called Transformation, acquired characteristics gained during an organism's life and can be inherited by the organism's offspring
- Charles Darwin – Father of Evolution, demonstrated that all living things evolved from earlier forms of life by the process of natural selection and published Origin of species
- Sigmund Freud – Founder of Psychoanalysis
 - importance of childhood experiences and sex in the development of affective disorder (Psychosexual Stages of Development)
 - structured model of ego psychology (id, ego, super-ego)
- Alan Turing – developed Turing Machine, basis for the first computer
 - Turing Test - a computer in one room that can communicate with humans in another room must be able to convince the humans that it is intelligent

2. S&T in the Philippines

2.1. Pre-Colonial Period

- Metal Age – significant influence in the lives of early Filipinos
- Developed tools in everyday life (planting, hunting, cooking, and fishing)
- Interpret movements of heavenly bodies for predicting seasons and climates.
- Made simple tools and weapons from stone. Which later developed by sawing and polishing.
- Discovered the use of medicinal plants.
- Weaved cotton, glass ornaments and cultivated lowlands and field terraces in mountain regions.
- Learned to build warfare boats (caracoa) which also used for trading purposes.
- Traded with China and Vietnam
- Counted years by the period of the moon and from one harvest to another
- With alphabet and writing system, method of counting and measurements.

2.2. Spanish Colonial Period

- Beginning of formal science and technology in the country.
- Spaniards established schools for boys and girls and introduced concept of subjects and disciplines.
- Established schools
 - Colegio de San Ildefonso – Cebu
 - Colegio de San Ignacio – Manila
 - Colegio de Nuestra Senora del Rosario – Manila
 - Colegio de San Jose - Manila
- Medicine and Pharmacy - Most developed science-based profession
 - University of Sto. Tomas – highest institution of learning
- Development of Hospitals
 - San Juan Lazaro Hospital – oldest in the far-east (1578)
- Manila Observatory - Founded by Jesuits to promote meteorological studies
 - Fr. Federico Faura issued the first public typhoon warning
- At the end of Spanish regime, the country had evolved into a primary agricultural exporting economy

2.3. American Period and Post-Commonwealth Era

- Science and technology in the Philippines advanced rapidly during the American regime due to:
 - extensive public education system
 - granting of scholarships for higher education in science and engineering
 - organization of science research agencies
 - establishment of science-based public services.
- Promoted higher education in sciences through University of the Philippines
 - Agriculture
 - Liberal Arts
 - Engineering
 - Veterinary Medicine
 - Law
 - Pharmacy
 - Education
- Replaced Bureau of Science with Institute of Science as primary research center
- Passed the Science Act of 1958
- Established the National Science Development Board
- Recognized National Research Council of the Philippines

2.4. Marcos Era and Martial Law

- Mandated Department of Education to promote science in public high schools
- Training and scholarships were awarded
- Presidential Decree No. 78 s. 1972 – established the Philippine Atmospheric, Geophysical and Astronomical Services (PAG-ASA)
- Other S&T Agencies established
 - Philippine National Oil Company
 - International Rice Research Institute
 - Philippine Coconut Research Institute
 - Philippine Council for Agriculture and Fisheries

2.5. Fifth Republic

- **Pres. Corazon Aquino**
 - National Science and Technology Authority (NSTA) was renamed as Department of Science and Technology
 - Crafted the Science and Technology Master Plan (STMP1991-2000) aiming to be a Newly Industrialized Country

- RA 6655 – Free Public Secondary Education Act of 1988
- **Pres. Fidel V. Ramos**
 - Significant increase in personnel specializing in S&T
 - Crafted the Science and Technology Agenda for Development (STAND) aiming to improve R&D in the country
 - Enacted laws for S&T during his term, including RA 8439, Magna Carta for S&T Personnel
- **Pres. Joseph Estrada**
 - Signed the RA 8749, or Clean Air Act of 1999 and RA 8792, or the Electronic Commerce Act
- **Pres. Gloria M. Arroyo**
 - Golden Age of Philippine S&T
 - RA 9637, Biofuel Act of 2006
 - To direct the use of biofuels, establishing for this purpose the biofuel program, appropriating funds therefore, and for other purposes.
- **Pres. Benigno Aquino III**
 - Recognized scientific researchers
 - Gavino C. Trono – seaweed species
 - Angel Alcala – marine biology research
 - Ramon Barba – induction of flowering of mango
 - Edgardo Gomez – national conservation of coral reefs
 - Abelardo Aguilar – discovered erythromycin
 - Gregorio Zara – invented videophone
 - Fabian Dayrit – research on herbal medicine
 - Diosdado Banatao – invented single-chip graphical use interface (for internet)
 - Daniel Dingel – water-powered car
 - Gregory Ligot Tangonan – research in the field of Communications Technology
 - Fe Del Mundo – invented incubator
 - Lourdes Cruz – research on sea snail venom

S&T in Nation Building

S&T Defined

- Science is the systematic process of obtaining knowledge of the physical and natural world through observation and experiment.
- Technology is the application of scientific knowledge for practical purposes.

Importance of S&T in National Development

- Wealth creation
- Improvement in the quality of life
- Society transformation
- Economic growth

Importance of S&T

- Better life
- Information
- Comfort
- Education
- Save time and money

Impact of Technology on Society

- Communication
- Information
- Transportation
- Education

The Role of S&T in Nation-Building

- One of the driving forces of economic growth (Research and development)
- The Role of S&T in the Developing World in the 21st Century

- Breakthrough in health services and education
- Improve infrastructure
- Determines socio-economic progress of a country

S&T Environment from 1986 to 2016

- Pres. Corazon Aquino highlighted the importance of S&T in achieving economic progress.
- Established the Department of Science and Technology (DOST) formulating the Science and Technology Master Plan (STMP1991-2000) aiming to be a Newly Industrialized Country
- Pres. Fidel V. Ramos introduced laws on Inventors' Incentives (RA 7459) and S&T Scholarships (RA 7687, RA 2067, RA 10612)
- Pres. Gloria Macapagal-Arroyo adopted policies focusing on a national innovation system and promoted technological entrepreneurship under Medium Term Plan (MTP 2004-2010).

D. Gov't Laws, Policies, Projects, and Plans

Government Policies and Laws Pertaining to S&T

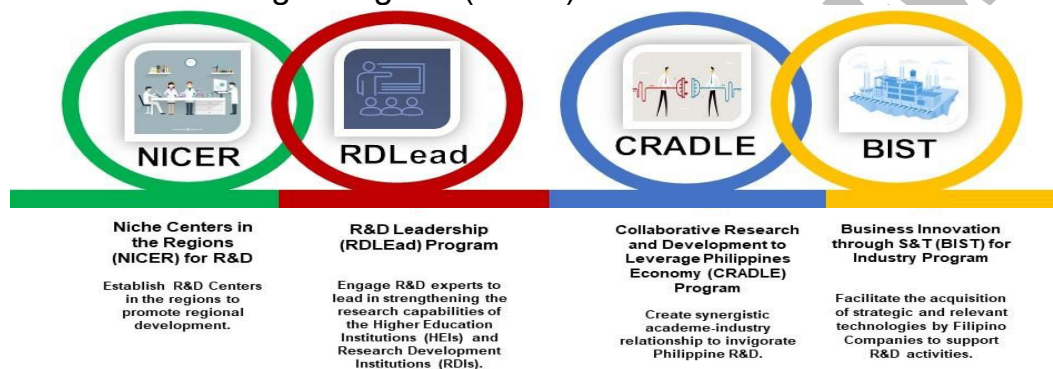
- RA 2067 - Science Act of 1958
 - To integrate, coordinate and intensify scientific and technological research and development and to foster invention to provide funds therefore and for other purposes.
- National Research Council of the Philippines
 - Act as the official adviser on scientific matters to the Government of the Philippines.
- RA 3589 – Act Amending RA 2067
 - No person may manufacture, produce, transfer, acquire, own, possess, import or export any radioactive material except in pursuance of a license issued in accordance with this Act.
- RA 6655 – Free Public Secondary Education Act
 - Opened free secondary education and promotes scientific and technological literacy through Science for the Masses Program
- RA 8439 - Magna Carta for Scientists Engineers Researchers and other S&T Personnel in the Government
 - To empower Filipino scientists, expand the country's pool of S&T experts, and motivate them to continue working for the government.
- RA 7459 - Inventors and Inventions Incentive Act
 - To empower Filipino scientists, expand the country's pool of S&T experts, and motivate them to continue working for the government.
- RA 8749 - Clean Air Act of 1999
 - To protect and preserve the environment and ensure the sustainable development of its natural resources.
- RA 8792 - Electronic Commerce Act
 - To outlaws computer hacking and provides opportunities for new businesses emerging from the internet-driven New Economy
- RA 9637, Biofuel Act of 2006
 - To direct the use of biofuels, establishing for this purpose the biofuel program, appropriating funds therefore, and for other purposes.
- RA 7687 - Science and Technology Scholarship Act of 1994
 - Provides for scholarships to talented and deserving students whose families' are under socio-economic status and do not exceed the set cut-off values of certain indicators.
- RA 2067 - DOST-SEI Merit Scholarship Program
 - Formerly known as NSTA Scholarship

- Awarded to students with high aptitude in science and mathematics and are willing to pursue S&T careers.
- RA 10612 - Fast-Track S&T Scholarship Act of 2013
 - Aims to strengthen the country's S&T education by fast-tracking graduates in the sciences, mathematics and engineering who shall teach science and mathematics in secondary schools throughout the country.
- RA 8293 - Intellectual Property Code
 - An Act prescribing the Intellectual Property Code and establishing the Intellectual Property Office, providing for its powers and functions, and for other purposes

Major S&T Development Plans

- S&T Master Plan (STMP) - shows the direction and desired goals of Philippine S&T for both the public and private sectors aiming the Philippines to become Newly Industrialized Country.
 - Sectoral Planning Councils
 - Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development (PCAARRD)
 - Philippine Council for Health Research and Development (PCHRD)
 - Philippine Council for Industry, Energy and Emerging Technology Research and Development (PCIEERD)
 - Research and Development Institutes
 - Advanced Science and Technology Institute (ASTI)
 - Food and Nutrition Research Institute (FNRI)
 - Forest Products Research and Development Institute (FPRDI)
 - Industrial Technology Development Institute (ITDI)
 - Metals Industry Research and Development Center (MIRDC)
 - Philippine Nuclear Research Institute (PNRI)
 - Philippine Textile Research Institute (PTRI)
 - Scientific and Technological Services
 - Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA)
 - Philippine Institute of Volcanology and Seismology (PHIVOLCS)
 - Philippine Science High School (PSHSS)
 - Science and Technology Information Institute (STII)
 - Science Education Institute (SEI)
 - Technology Application and Promotion Institute (TAPI)
 - Collegial and Scientific Research Bodies
 - National Academy of Science and Technology (NAST)
 - National Research Council of the Philippines (NRCP)
 - Major S&T Development Plans
- S&T Agenda for National Development (STAND Philippines, 2000)
 - Emphasizes the development and utilization of superior technologies to a level of competitive advantage.
 - Identified investment areas
 - export winners, domestic needs, supporting industries and coconut industries
- National S&T Plan (NSTP 2002-2020)
 - By 2004, contributed significantly to the enhancement of national productivity
 - By 2010, earned niches and become a world class knowledge provider and user in selected S&T area
 - BY 2020, developed a wide range of globally competitive products and services of high technological content.
 - Priority Areas for S&T Development in NSTP 2002-2020
 - Agriculture, Forestry and Natural Resources
 - Health and Medical Services

- Biotechnology
- ICT
- Microelectronics
- Materials, Science and Engineering
- Earth and marine services
- Fisheries and aquaculture
- Environment
- Natural Disaster mitigation
- Energy
- Manufacturing and service engineering
- National Integrated Basic Research Agenda, NIBRA 2022-2028
 - Blue Skies (Pure Basic Research)
 - Issue-Based (Oriented Basic Research)
 - Tubig: Water Security (Tubig Ay Buhayin At Ingatan)
 - Alert: Clean Energy (Alternative Energy Research Trends)
 - Likas: Health Sufficiency (Likas Yaman Sa Kalusugan)
 - Atin: Inclusive Nation-Building(Ang Tinig Natin)
 - Sapat: Food And Nutrition Security (Saganang Pagkain Para Sa Lahat)
 - Saklaw: Sustainable Community (Saklolo Sa Lawa)
 - Policy Research
- Science for Change Program (S4CP)



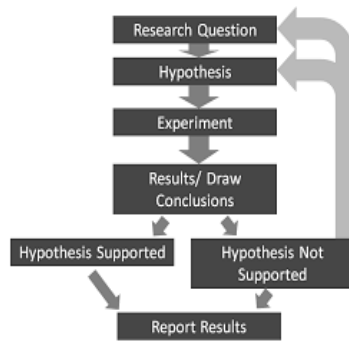
- Status of S&T and Plans for Development under President Rodrigo Duterte
 - DIWATA-1 satellite, made by Filipino scientists was launched in 2016
 - DIWATA-2 satellite, was improved and was launched in 2018
 - Installed the Zamboanga Doppler Weather Radar System
 - RA 11035 – Balik Scientist Program - To give more incentives to returning Filipino experts, scientists, inventors and engineers who would share expertise in the country.

B. STS and the Human Condition

B.1. The Human Flourishing

B.1.1. Aristotle Asks

- Eudaimonia “good spirited” - To describe the peak of human’s happiness that can be attained.
- Flourishing is the highest good of human attempt to achieve his/her goal. It is a success as a human being when he/she achieved it for the betterment of his/her life
 - Western civilization tends to focused on the individual (individual flourishing).
 - Eastern civilization tends to put community above the individual (community-centric).



- Science as a Method

B.1.2. Science as a Social Endeavor

- Balance possible biases
- Stimulate problem solving
- Facilitate specialization
- Inspire and motivate
- Divide labor

B.1.3. What are the Human Conditions before and after science and technology?

Technological advancements and their effects on Humanity (Karehka Ramey, 2012)

1. Transportation
2. Communication
3. Business
4. Medicine and health
5. Agriculture
6. Education

B. 2. The Good Life

B.2.1. What is meant by a good life?

- Moral approval (Socrates)
- Pleasures in life (Epicurus)
- Fulfilled life (Aristotle)
- Meaningful life

B.2.2. Materialism, Hedonism, Stoicism, Theism, Humanism

- Materialism – comfort, pleasure and wealth are the only highest goals.
 - Leucippus and Democritus
 - Atomist Philosophers, universe and matter are only made up of atoms
- Hedonism – see the end goal of life in attaining pleasure
 - Epicurus, life is limited, one must indulge itself with pleasures
 - “Eat, drink and be merry for tomorrow we die”
- Stoicism – asserts virtue is happiness
 - Marcus Aurelius
 - People do not have any control must not rely on external events
- Theism – one or more Gods exist within universe that are omniscient (all-knowing), omnipotent (all-powerful) and omnipresent (all-pervasive).
 - Monotheism
 - Polytheism
- Humanism – human beings have right and responsibility to give meaning and shape their own lives.
 - Stewards of creation
 - Have control for themselves and world outside them.

B.2.3. What is the ultimate goal of life?

- Eudaimonia True happiness is found by leading a virtuous life and doing what is worth doing.
- Realizing human potential is the ultimate goal.

B.3. When Technology and Humanity Cross

- Television Sets, Mobile Phones, Computers, and Humanity
 - Paul Gottlieb Nipkow, send images through wires with the aid of rotating metal disk, “electric telescope”
 - Alan Archibald Campbell Swinton and Boris Rosing, used cathode-ray tube in television, mechanical and electronic television
 - Martin Cooper, made the first mobile phone, Motorola
 - Charles Babbage, originated the concept of a digital programmable computer
 - Osborne 1, first true portable computer, Laptop
- Roles Played by These Technological Advancements
 - Advertisements and information dissemination
 - Recreational activity
 - Propagandas and advocacies
 - Communication
- Robotics and Humanity
 - Robot is actuated mechanism programmable in two or more axes with degree of autonomy, moving within its environment, and performed intended tasks.
 - A service robot perform useful tasks for humans or equipment excluding industrial automation application.
 - A personal robot is a service robot for non-commercial tasks
 - Roles Played by Robot
 - Autonomous Mobile Robots (AMRs) - make decisions in near real-time as they go.
 - Automated Guided Vehicles (AGVs) - tracks or predefined paths and often require operator oversight.
 - Articulated Robots - emulate the functions of a human arm.
 - Humanoids - perform human-centric functions and often take human-like forms.
 - Cobots - function alongside or directly with humans.
 - Hybrids - capable of more complex tasks
 - Ethical Dilemma/s Faced by Robotics
 - Privacy & Surveillance - information privacy and the right to secrecy and personhood
 - Manipulation of Behavior - vulnerability to “nudges”, manipulation, and deception
 - Opacity of AI Systems - lack of due process, accountability, community engagement, and auditing
 - Bias in Decision Systems - where and when police forces will be needed most
 - Human-Robot Interaction – can not be replaced by robots: care, love, and sex.
 - Ethical Dilemma/s Faced by Robotics
 - Automation and Employment - “job polarization”
 - Autonomous Systems - must adapt to or whether they just require technical adjustments.
 - Machine Ethics - ensuring that the behavior of machines toward human users and other machines
 - Artificial Moral Agents – robots having rights and responsibilities
 - Singularity - trajectory of artificial intelligence reaches up to systems that have a human level of intelligence

C. Specific Issues in STS

C.1. Information Age

C.1.1. Historical Development

- 21st century, also called Computer Age
- Characterized by rapid shift from traditional industry

- 4 Periods
 - Pre-Mechanical – 3,000 BC to 1450 AD
 - Mechanical – 1450-1840
 - Electro-Mechanical – 1840-1940
 - Electronic/Information – 1940 to present
- Clade Shannon, Father of Information Age

C.1.2. Living in the Information Age

- Advantages of Information Technology
- Globalization
- Communication
- Cos-effectiveness
- Creation of new jobs

C.2. Understanding Gene Therapy

C.2.1. Understanding Gene Therapy

- Gene therapy is a technique that uses a gene(s) to treat, prevent or cure a disease or medical disorder.
 - adding new copies of a gene that is broken
 - replacing a defective or missing gene in a patient's cells with a healthy version of that gene
- Ex vivo - introducing normal genes in isolated cells from patients
- In vivo - introducing DNA directly into cells while they are in the patient.
- Genetic Engineering - It is a process to alter the DNA makeup of an organism by changing a single base pair (A-T or C-G), deleting a region of DNA or adding a new segment of DNA.

C.2.2. Diseased People Cured by Fixing their Genes

- CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) to treat sickle cell anemia
- CAR-T cell therapy (Chimeric antigen receptor) to treat acute lymphocytic leukemia (ALL)
- Luktuna, FDA approved voretigene neparvovecrzyl to treat retinal dystrophy
- Spinraza, FDA approved nusinersen to treat Spinal muscular atrophy
- Diseased People Cured by Fixing their Genes

C.2.3. Ethical Considerations

- Justice and equity
- Informed consent
- Genome-Editing Research Involving Embryos
- Safety

C.2.4. The Role of Genes in Heredity

- Gene is considered the basic unit of inheritance.
- Humans have approximately 20,000 protein-coding genes.

C.2.5. Effects of Gene in our Environment

- Major risks of GMOs
 - Risk in gene flow
 - Emergence of superweeds
 - Recombination of viruses and bacteria to produce new pathogens
- Direct risks of GMOs
 - Disruption of natural environment (competition and interference)
 - Unexpected behavior of GMO upon escape
 - Interfere natural biochemical cycles
 - Negative impact to consumer
- DNA Code

- Genetic code is the set of instructions in a gene that tell the cell how to make a specific protein.

C.3.1. Biodiversity and Healthy Society

C.3.1. The impact of Biodiversity on Human Life

- Nutrition
- Human health
- Environment-related illnesses

C.3.2. Human Population Growth

- 6th Mass Extinction
 - The reduction in biodiversity since the emergence of humans is now on the scale of another mass extinction

C.3.3. What are the top ten environmental problems in the world

- Pollution
- Global warming
- Overpopulation
- Waste disposal
- Ocean acidification
- Loss of biodiversity
- Deforestation
- Ozone layer depletion
- Acid rain
- Public health issues

C.4. Nanotechnology

- Nanotechnology is science, engineering, and technology conducted at the nanoscale (1 to 100 nm)
- Richard Feynman, Father of Nanotechnology
- Commercial Applications
 - Body armor/vests
 - Surface protection materials
 - Solar panels
 - Food products and packaging
 - Transdermal patches
 - Bandages
- Potential Uses
 - Energy production
 - Agriculture productivity
 - Water treatment and remediation
 - Disease screening and diagnosis
 - Health monitoring
 - Food processing and storing
 - Air pollution control
 - Construction
 - Pest detection and control

D. Climate Change and Energy Crisis

- Climate change refers to long-term shifts in temperatures and weather patterns.
- Causes of Climate Change
 - Natural Causes
 - Volcanic eruptions
 - Orbital Changes
 - Variation in Solar Radiation
 - Movement of Crustal Plates
 - El Niño-Southern Oscillation (ENSO)

- Changes in Greenhouse Gases
 - Sources of Greenhouse Gases
 - Carbon dioxide
 - Water vapor
 - Nitrous oxide
 - Methane
 - Sources of Carbon dioxide
 - Respiration and decomposition
 - Weathering of carbonate rocks
 - Burning of fossil fuels and forests
- How is Climate Change affecting the Philippines?
 - More intense El Nino
 - Sea surface temperatures to rise
 - Ocean acidification
 - Sea levels rise by 4 to 6 meters
 - Tropical cyclones
 - Rainfall, river flow and flooding
 - Altering agricultural production
 - Reduction in crop yields
 - Food security declines
 - Malnutrition increases
- Energy Crisis - It is the foreseeable end of the cycle of oil, gas and coal which takes hundreds of thousands of years to replenish.
- Causes of Energy Crisis
 - Overconsumption
 - Overpopulation
 - Poor infrastructure
 - Unexplored renewable energy options
 - Delay in Commissioning of Power Plants
 - Wastage of energy
 - Poor distribution system
 - Major accidents and natural calamities
 - Wars and Attacks
 - Miscellaneous Factors
- Possible solutions to Energy Crisis
 - Move towards renewable resources
 - Buy energy efficient products
 - Lighting controls
 - Easier Grid Access
 - Energy Simulation
 - Perform Energy Audit
 - Common Stand on Climate Change

E. Environmental Awareness

E.1. Promotion of Environmental Awareness

- Components of Environmental Education
 - Awareness and sensitivity
 - Knowledge and understanding of the environment
 - Attitude of concern and motivation to improve or maintain environment quality
 - Skills to identify and help resolve
 - Participation in activities that lead to resolution

E.2. Top 10 benefits of Environmental Education

- Heightened imagination and enthusiasm.
- Learning transcends the classroom
- Critical and creative thinking are enhanced

- Develop tolerance and understanding
- State and national learning standards are met for multiple subjects
- Decline in biophobia and nature deficit disorder
- Healthy lifestyle
- Communities are strengthened
- Responsible action is taken to better the environment
- Students and teacher are empowered.

E.3. Beliefs and Convictions

- Environmental Stewardship
 - Help in forging community linkages for nature preservation activities
 - Educate people on proper waste segregation
 - Practice 7R
 - Save energy

F. Alternative Energy Resources

- What is Alternative Energy?
 - It refers to energy sources other than fossil fuels (coal, gasoline, and natural gas) including all renewable resources and nuclear power.
 - Intended to address concerns of high carbon emissions.
- Differences between Alternative Energy and Renewable Energy
 - **Renewable resources** comes from sources that are **naturally occurring** and **replenishes**.
- Sources of Alternative Energy
 - Nuclear power
 - Wind power
 - Solar power
 - Hydraulic power
 - Geothermal power
 - Space-based solar power
 - Tidal power
 - Biofuel
 - Hydrogen or hydrosulfuric power
- Alternative and Renewable Energy Sources in the Philippines
 - Hydroelectric Plants
 - Angat Dam, in Bulacan is the major hydropower facility in the Philippines
 - There are 14 conventional dams and 15 run-of-the-river systems
 - Alternative and Renewable Energy Sources in the Philippines
 - Geothermal Plants
 - Suitable for areas with low winds, such as Mindanao, and areas that have rainy weather, such as Batanes.
 - Solar Power
 - The country is geographically located in a region that receives a high amount of sunlight each year.
 - Wind Power
 - All wind power sites in the Philippines are on-shore facilities and are tourist destinations
 - Biomass Power
 - Refers to energy derived from plant and animal sources.
 - Bagasse, rice husks, coconut husks, as well as landfill are used to generate power.